

Kaustubh Khulbe

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EDUCATION

University of Illinois Urbana-Champaign

May 2026

Bachelor of Science, Computer Science

GPA 4.00/4.00

- **Relevant Coursework:** Advanced Operating Systems, Advanced Comp Arch. - Mobile and Edge Computing, Operating Systems, Applied Parallel Programming with CUDA, Comp Arch., Algorithms, Data Structures
- **Student Organizations:** IEEE, Illinois Boxing Club
- **Awards:** Illinois Engineering Achievement Scholar, Dean's List

PROFESSIONAL EXPERIENCE

University of Illinois Urbana-Champaign

December 2024 - Present

ECE 391 (Operating Systems) Course Assistant

Urbana, IL

- Building a robust auto-grader to allow automatic interfacing with QEMU and testing student software
- Hosting office hours for a class size of 200+ students
- Expanding final OS project by integrating user-space threads, shared memory, and a pre-emptive priority scheduler

Undergraduate Researcher

August 2024 - Present

UIUC ADAPT Lab

Urbana, IL

- Developing a tensor framework/compiler with high-level ISA guidelines for accelerators
- Describing the Berkeley GeneSys accelerator architecture with a custom language and XLA
- Benchmarked distributed transformer mechanisms like FSDP across multi-GPU clusters

Software Engineer Intern

May 2024 - August 2024

Omni RPA Inc.

San Jose, CA

- Built a network transfer cost optimizer for an orchestrator using Go and Nebula, saving thousands of dollars
- Designed large-scale graph models to represent pricing and performance models in Snowflake and Nebula DB
- Constructed scalable microservices to automate downloading, parsing, and uploading cloud provider data

Software Engineer Intern

March 2024 - May 2024

Kind Kart

San Jose, CA

- Worked with a non-profit that raised over \$124,000 to streamline donations using GraphQL, PostgreSQL, and Go
- Automated services with Cron jobs to update donation statistics and send emails to clients
- Compiled documentation on testing procedures which helped onboarding and wrote numerous integration tests

PROJECTS

Illinux: Operating System | C, Unix, RISC-V, QEMU

July 2024 - December 2024

- RISC-V Unix-like OS with paging, filesystem, devices, system calls, and a pre-emptive round robin scheduling
- Developed a virtio block driver, cryptography driver, and entropy driver
- Kernel supports pipes between processes and argument passing according to POSIX standards
- **2nd place** in OS design competition in ECE department with 100+ teams

CUDA-based LeNet CNN Network | C, C++, CUDA, nsys, ncui

August 2024 - December 2024

- Developed and optimized forward pass of LeNet CNN Network
- Designed and benchmarked with modern accelerator features such as streams and tensor cores using nsys and ncui
- Achieved lossless speedups of 70.45% from base trivial GPU implementation
- Used cuBLAS, half precision floating points, and constant memory on A100 GPUs to improve cache and SM usage

Rust-Based DNS Server | Rust

May 2024 - June 2024

- Built a custom thread-safe DNS server that handled the DNS protocol for queries and responses using Rust
- Improved performance of query times from 81.99ms to 0.24ms by building an LRU cache and multithreading
- Added safety to the system through logging, error handling, and backing up server states

SKILLS

Programming Languages: C, C++, Rust, Python, Go, MIPS ISA, RISC-V ISA, Verilog, SystemVerilog

Tools and Technologies: Git, Linux, QEMU, CMake, gdb, g++, nsys, ncui

Frameworks and Libraries: CUDA, Pytorch, virtio